Databases Report

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1 Database Design

Our objective was to build a database that ensures data integrity and query efficiency. The data was split into six entities: dimension tables (`Country` and `Time`) are connected to fact tables (`Covid_Statistics` and `Vaccination_Statistics`), and enriched by static profile tables (`Country_Profile` and `Country_Age_Structure`).

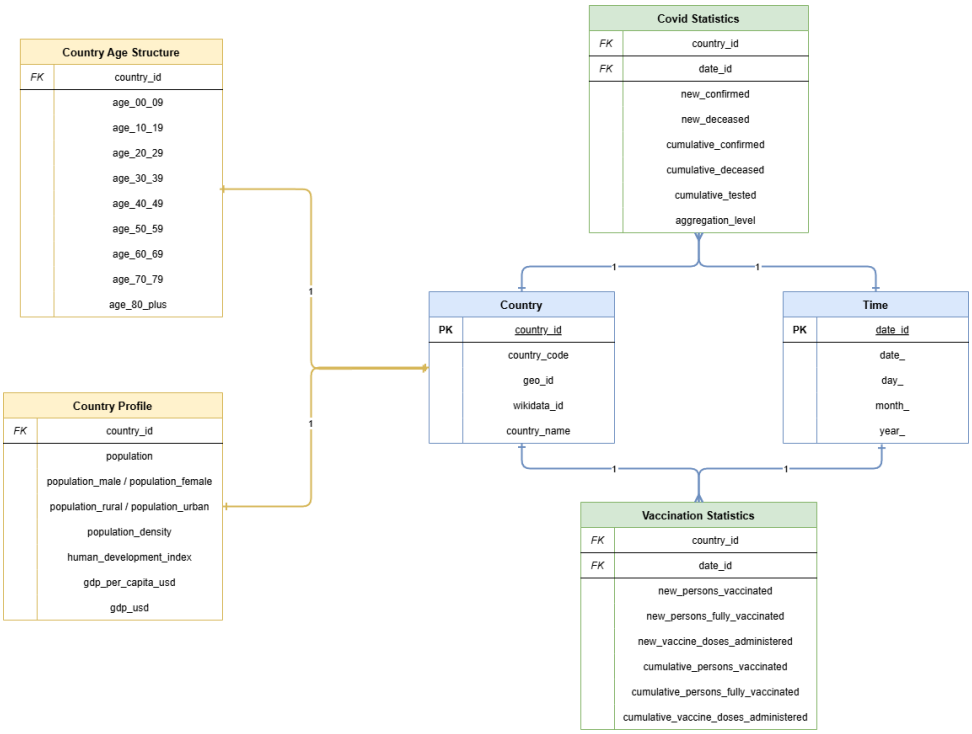

Figure 1: Entity-Relationship diagram of the database.

Country

The `Country` table contains the list of countries present in the datasets. Each country is uniquely identified and associated with multiple identifiers used in the original sources. By storing country attributes in a dedicated dimension table, the database ensures efficient joins with fact tables.

- `COUNTRY_ID` – surrogate primary key generated in the database.
- `COUNTRY_CODE` – country code used in the datasets.
- `GEO_ID` – geographical identifier from the European dataset.
- `WIKIDATA_ID` – identifier linking the country to Wikidata.
- `COUNTRY_NAME` – full country name.

This table acts as a dimension table connected to all fact tables containing country-level statistics.

Time

The `Time` table represents the temporal dimension of the database. COVID statistics and vaccination data are recorded daily; therefore, a dedicated time dimension is required to enable temporal analysis such as trend monitoring and monthly aggregation.

- `DATE_ID` – surrogate primary key.
- `DATE_` – the actual calendar date.
- `DAY_` – day of the month.
- `MONTH_` – month number.
- `YEAR_` – year.

Using a time dimension simplifies queries and enables efficient analytical operations in the data warehouse.

Covid Statistics

The `Covid_Statistics` table stores the daily evolution of the pandemic for each country. Each record corresponds to one country on one date and includes indicators describing the spread and severity of the disease. This table is a fact table referencing both the `Country` and `Time` dimensions.

- `COUNTRY_ID` – foreign key referencing the `Country` dimension.
- `DATE_ID` – foreign key referencing the `Time` dimension.
- `NEW_CONFIRMED` – number of new confirmed cases on that date.
- `NEW_DECEASED` – number of new deaths on that date.
- `CUMULATIVE_CONFIRMED` – total confirmed cases up to that date.
- `CUMULATIVE_DECEASED` – total deaths up to that date.
- `CUMULATIVE_TESTED` – total number of tests performed.
- `AGGREGATION_LEVEL` – level of geographical aggregation of the record.

The unique constraint on `(COUNTRY_ID, DATE_ID)` ensures that each country has exactly one record per day.

Vaccination Statistics

The `Vaccination_Statistics` table records the daily progress of vaccination campaigns for each country, allowing the analysis of rollout speed and its evolution over time. Like the `Covid_Statistics` table, it is a fact table linked to the `Country` and `Time` dimensions.

- `NEW_PERSONS_VACCINATED` – number of individuals receiving their first vaccine dose on that day.
- `CUMULATIVE_PERSONS_VACCINATED` – total number of individuals having received at least one dose.
- `NEW_PERSONS_FULLY_VACCINATED` – number of people completing their vaccination on that day.
- `CUMULATIVE_PERSONS_FULLY_VACCINATED` – total number of fully vaccinated individuals.
- `NEW_VACCINE_DOSES_ADMINISTERED` – number of doses administered that day.
- `CUMULATIVE_VACCINE_DOSES_ADMINISTERED` – total number of doses administered.

The `(COUNTRY_ID, DATE_ID)` constraint ensures the uniqueness of daily vaccination records.

Country Profile

The `Country_Profile` table stores static demographic and socioeconomic indicators for each country. Unlike the fact tables, these attributes do not change daily and serve to contextualise differences in pandemic impact across countries.

- `POPULATION` – total population.
- `POPULATION_MALE` / `POPULATION_FEMALE` – male and female population counts.
- `POPULATION_RURAL` / `POPULATION_URBAN` – rural and urban population counts.
- `POPULATION_DENSITY` – population density (inhabitants per km^2).
- `HUMAN_DEVELOPMENT_INDEX` – HDI indicator.
- `GDP_PER_CAPITA_USD` – GDP per capita in US dollars.

This table is linked to the `Country` dimension through `COUNTRY_ID`.

Country Age Structure

The `Country_Age_Structure` table describes the age distribution of each country's population. Age distribution is a critical factor in COVID-19 analysis, since the disease has markedly different impacts across age groups.

- `AGE_00_09`, `AGE_10_19`, `AGE_20_29`
- `AGE_30_39`, `AGE_40_49`, `AGE_50_59`
- `AGE_60_69`, `AGE_70_79`, `AGE_80_PLUS`

Each record corresponds to one country and is linked through the `COUNTRY_ID` primary key.

Importation of the data

Once the database had been created in Oracle, the data was imported into it and prepared for analysis. This stage was essential, as the dataset was particularly large, with more than 12 million rows in total.

Because of this volume, importing the data into Oracle required careful handling in order to ensure that the process remained stable and that the data was loaded correctly. After the importation phase, the database was fully operational and could be used to run the different SQL queries presented in this report.

2 Queries on the database

2.1 Query 1

Firstly, we wanted to study the Fatality Rate of each country according to their GDP per Capita One of the most intuitive hypotheses is that wealthier countries, benefit from stronger healthcare system and would exhibit lower COVID-19 fatality rates.

To investigate this, we queried the database to get each country's GDP per capita and its overall case fatality rate (CFR), defined as the ratio of total confirmed deaths to total confirmed cases expressed as a percentage.

The results, presented in Table 1, reveal a nuanced picture. Among the wealthiest nations in the dataset — Monaco, Liechtenstein, Luxembourg, and Switzerland — fatality rates are indeed low, generally below 0.5%. However, the relationship is not uniform: Belgium and the United Kingdom, high-income countries, have higher fatality rates. It suggests that wealth alone does not explain these outcomes.

Brazil's GDP figure (2,490,617) looks unusually high, with a fatality rate of 1.98.

Table 1: Fatality Rate by Country and GDP per Capita

Country	**GDP per Capita (USD)**	**Fatality Rate (%)**
Brazil	2,490,617	1.98
Monaco	185,829	0.44
Liechtenstein	173,356	0.44
Luxembourg	116,348	0.39
Cayman Islands	85,477	0.10
Switzerland	81,993	0.33
Belgium	81,892	0.72
Isle of Man	80,989	0.31
Ireland	78,706	0.48
Germany	76,464	0.46
United Kingdom	75,756	0.80

2.2 Query 2

Secondly, we wanted to study the Fatality Rate across the months of the year. The hypothesis is that COVID-19 mortality would follow a seasonal pattern.

To investigate this, we queried the database to retrieve for each month the total number of confirmed cases, total deaths, and the resulting mortality rate.

The results, presented in Table 2 show that Fatality rates peak in May (1.78%) and April (1.44%), while summer months show lower rates.

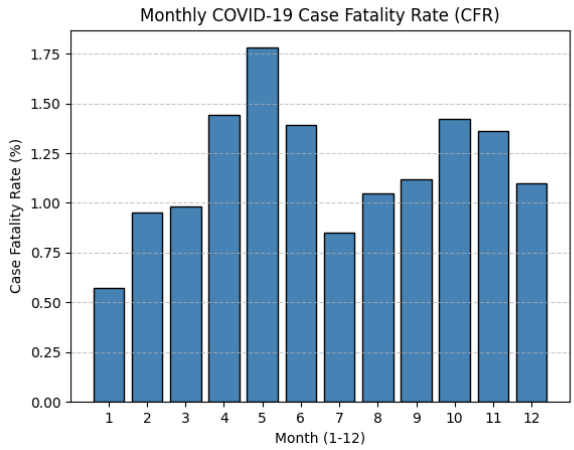

Figure 2: Monthly COVID-19 Mortality Rate

2.3 Query 3

We also studied whether colder climates were associated with more intense COVID-19 transmission. The hypothesis is that lower average temperatures favour the spread of the virus, as cold weather drives populations indoors, increases close contact... To measure spread intensity independently of country size, we used the cumulative confirmed cases per 100,000 inhabitants as the primary indicator, alongside the raw total case count for comparison.

To investigate this, we queried the database to retrieve for each country its average temperature and its cumulative infection rate per 100,000 inhabitants, restricting the analysis to the coldest countries in the dataset.

The results, presented in Figure 3, reveal two contrasting pictures depending on the metric used. The left panel, showing raw total confirmed cases, appears to suggest no clear relationship between cold temperatures and case volume. Most cold countries have relatively low absolute case counts, with the exception of one outlier near 5°C recording over 20 million cases, likely reflecting a large and well-tested populous country. This metric is heavily biased by population size and testing capacity, making it a poor indicator of actual spread intensity.

The right panel, normalised per 100,000 inhabitants, tells a more nuanced story. Countries clustered between 0°C and 10°C exhibit the highest infection rates, with several exceeding 50,000 cases per 100,000 inhabitants. Meaning more than half the population was officially recorded as infected at some point. Interestingly, the single country at −30°C shows a moderate infection rate of around 21,000 per 100,000, suggesting that extreme cold and likely extreme isolation may actually limit transmission despite the unfavourable climate. Overall, the results provide partial support for the hypothesis: moderately cold and densely populated countries appear most affected, while the relationship breaks down at temperature extremes.

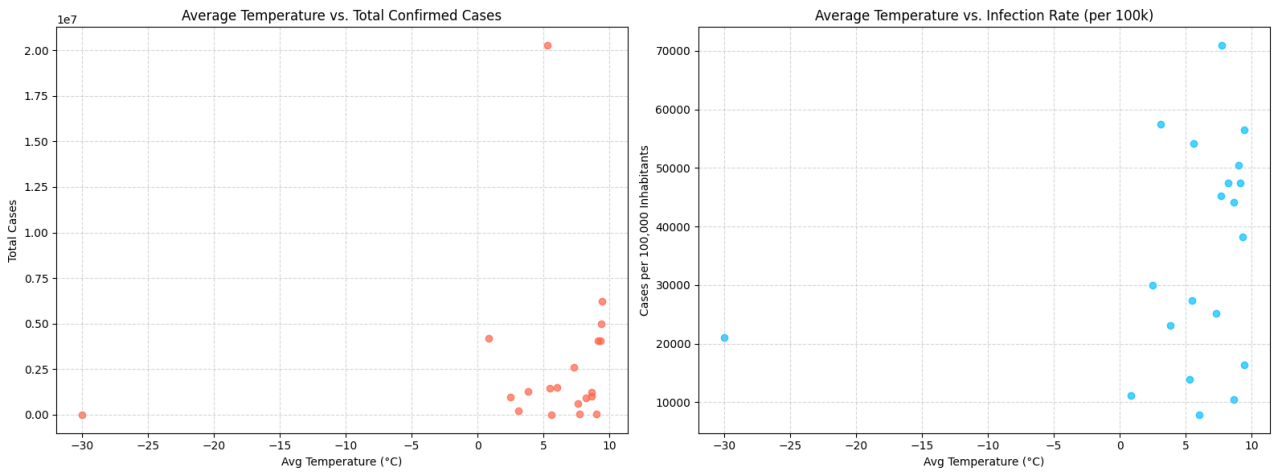

Figure 3: Average country temperature versus total confirmed cases (left) and infection rate per 100,000 inhabitants (right).

2.4 Query 4

We investigated the impact of vaccination on the apparent case fatality rate (CFR) in France over time. The hypothesis is that as vaccination coverage increased progressively throughout 2021, the monthly CFR should exhibit a clear decline, reflecting the protective effect of vaccines.

To investigate this, we queried the database to compute for France, on a monthly basis: the total number of new confirmed cases, the total number of new deaths, the resulting apparent case fatality rate (CFR), and the cumulative number of vaccinated individuals.

The results, presented in Figure 4, reveal a striking and rapid evolution of the CFR over the pandemic period. In the early months of 2020, the CFR reached extreme values, peaking at 26.5% in April 2020. From mid-2020 onwards, as testing expanded significantly, the CFR dropped to below 2%, providing a far more realistic picture of the disease's lethality in the general population. Following the beginning of the vaccination campaign in early 2021, the CFR continued its decline and stabilised at very low levels (below 0.5%) throughout 2022, consistent with the hypothesis that vaccination reduced the likelihood of confirmed cases resulting in death.

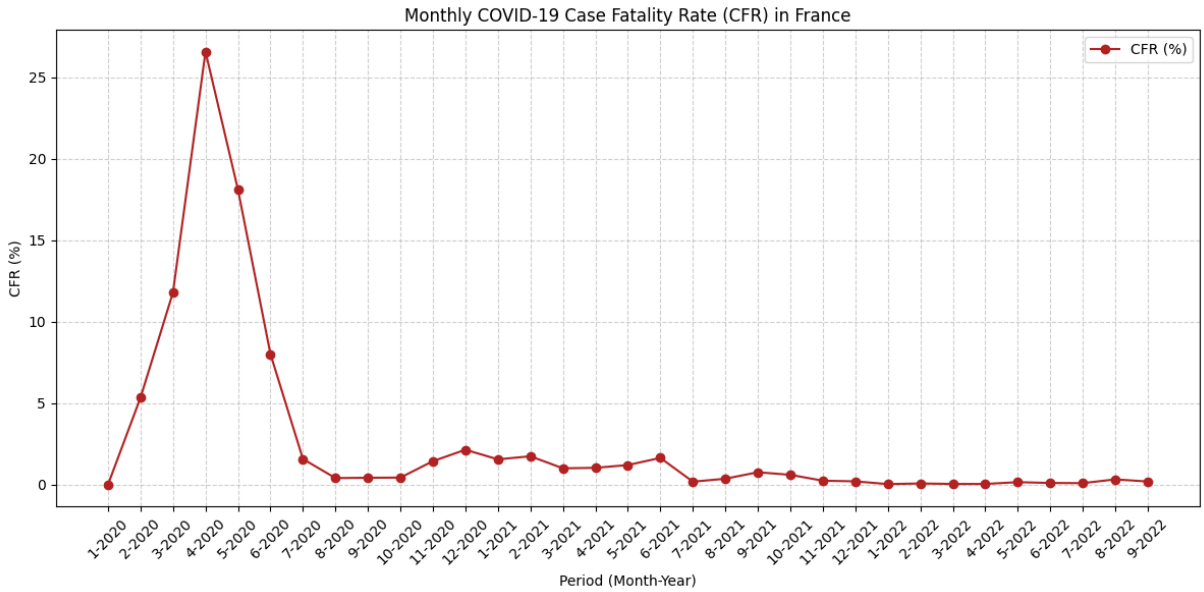

Figure 4: Monthly evolution of the apparent COVID-19 Case Fatality Rate (CFR) in France from January 2020 to September 2022.

2.5 Query 5

We investigated the vaccination impact threshold for the five largest European countries (France, Germany, Italy, United Kingdom, and Spain). The central question is: at what vaccination rate did weekly deaths first begin to decline in each country during the 2021–2022 period?

The underlying hypothesis is that once a sufficient share of the population is vaccinated, we should observe a drop in weekly deaths thanks to the immunity effects and reduced severe illness.

To investigate this, we queried the database to identify, for each country, the first week in which weekly deaths fell below the previous week, and recorded the cumulative vaccination rate at that moment, expressed as a percentage of the total population.

The results, presented in Figure 5, show differences across countries. France and Germany saw their first weekly death decline at very low vaccination rates (0.69% and 1.45% respectively), suggesting these early declines may reflect other dynamics rather than a vaccine effect.

Italy follows at 2.25%. By contrast, the United Kingdom (8.79%) and Spain (18.17%) reached their inflection point at considerably higher vaccination rates, which may indicate that their first measurable decline was more robustly linked to vaccine coverage.

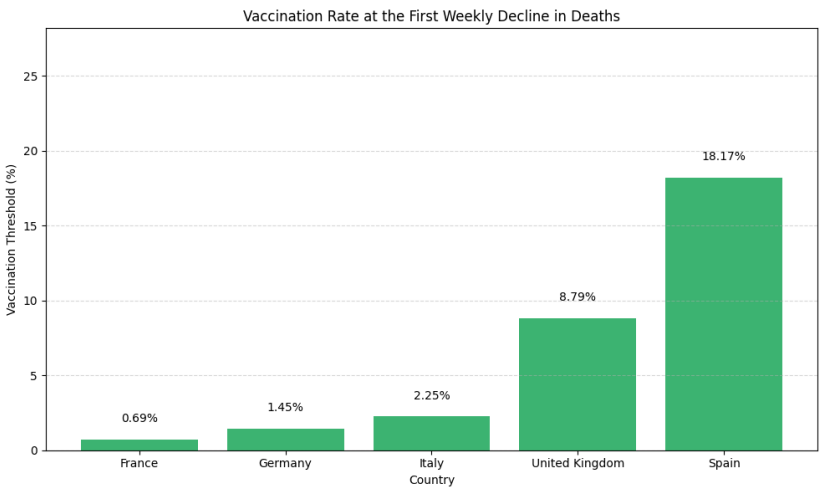

Figure 5: Vaccination rate at the first weekly decline in deaths for the five largest European countries (2021–2022).

2.6 Query 6

We also investigated whether countries with a larger elderly population share tend to suffer higher COVID-19 fatality rates. The hypothesis is: since COVID-19 is known to affect older individuals, countries with an older demographic structure should exhibit higher case fatality rates. The analysis was restricted to countries with a population above 10 million.

To investigate this, we queried the database to retrieve, for each qualifying country, the share of the population aged 70 and over alongside its overall case fatality rate (CFR).

The results, presented in Figure 6, paint a more complex picture than the hypothesis suggests. Overall, no clear linear relationship emerges from the scatter plot.

While some highly aged countries such as Italy (17.0%, CFR 0.80%) and Greece (16.5%, CFR 0.69%) show moderate fatality rates, Japan stands out as a remarkable outlier: despite having the highest elderly share in the dataset (21.8%), it recorded one of the lowest fatality rates (0.23%). It likely shows its strong healthcare system and effective containment.

Conversely, Ukraine and Romania, with comparatively lower elderly shares (around 11–12%), exhibit the highest fatality rates in the dataset (2.13% and 2.06%)

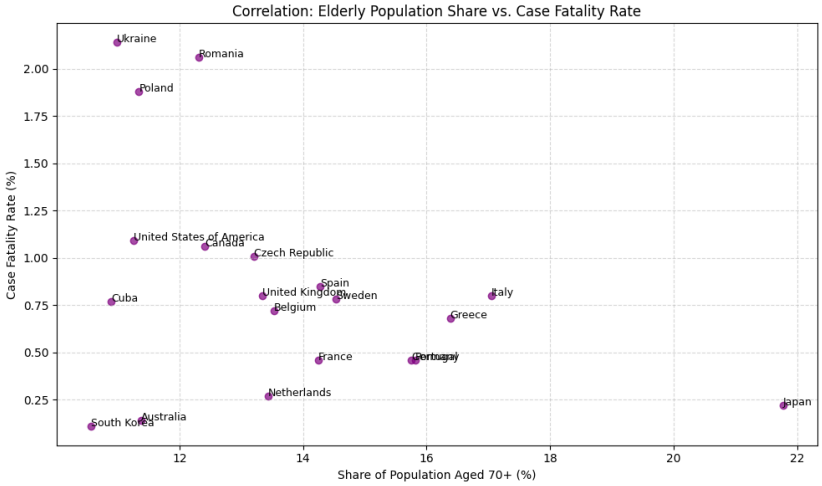

Figure 6: Correlation between the share of population aged 70+ and the case fatality rate (%) for countries with population above 10 million.

2.7 Query 7

We hypothesized that countries with a higher density of medical professionals—specifically physicians and nurses—would be better equipped to provide intensive care, thereby reducing the overall Case Fatality Rate (CFR). To test this, we performed a relational join between the country-specific healthcare capacity metrics and the aggregated mortality data. The density is measured as the number of professionals per 1,000 inhabitants.

The results, visualized in Figure 7, illustrate the correlation for both physicians (blue) and nurses (red). Between 0 and 2 physicians per 1,000 people, there is a high volatility in mortality, with several countries exceeding a 4% CFR. As the density of physicians increases beyond 3.0 per 1,000, the CFR appears to stabilize and remain consistently below 2%. The nurse density plot shows an even more pronounced trend. Some outliers exist in the physician data, suggesting that during peak surges, even well-staffed systems may reach a saturation point where marginal gains in staff density diminish.

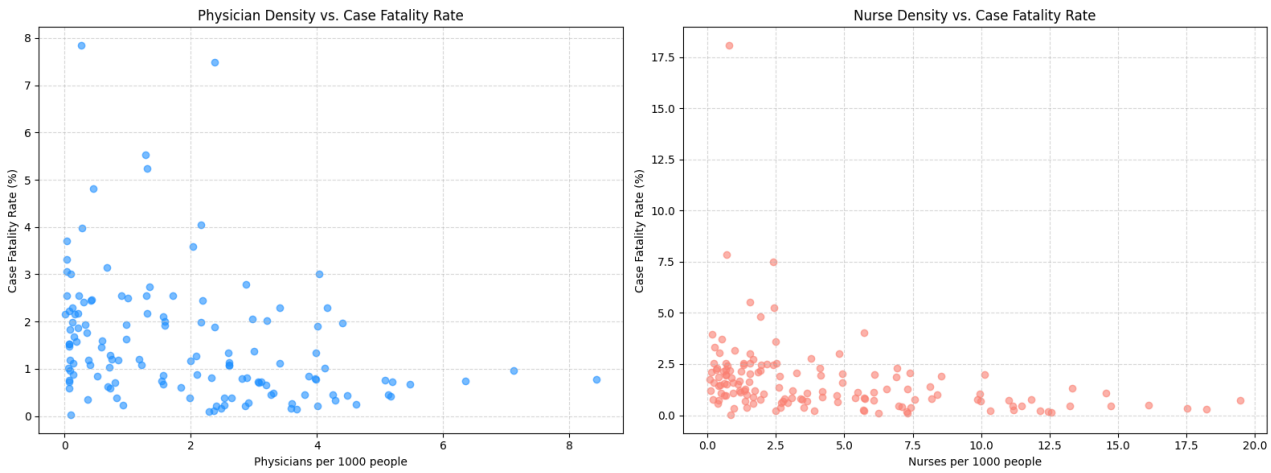

Figure 7: Correlation between Physician and Nurse density (per 1,000 people) and the Case Fatality Rate (%).

2.8 Query 8

Our final query focused on France and aimed to identify the months during which the country faced the most intense pandemic pressure. Rather than relying on a single indicator, which can be misleading in isolation, we constructed a composite *Pandemic Severity Score* combining three dimensions: the volume of new confirmed cases, the total death toll, and the apparent case fatality rate.

The results, presented in Figure 8, tell a clear and compelling story across two complementary panels.

The upper panel shows the monthly severity score over the entire pandemic period. A dominant peak emerges in January 2022, where the score reaches approximately 1.05×10^9, far exceeding all other months. This period corresponds to the Omicron wave, which drove a volume of confirmed cases in France. The months of February, March, and April 2022 also register elevated scores, indicating that this wave was sustained over several months rather than a brief spike. Earlier waves in late 2020 and throughout 2021 produced moderate severity scores, while the initial wave of Spring 2020, despite its dramatic CFR, generated a comparatively low composite score due to the extremely limited case detection at the time.

The lower panel overlays the monthly CFR (red) against the cumulative vaccination progress (green). Despite the Omicron wave generating the highest severity score on record, the CFR during that period remained extremely low (below 0.5%), demonstrating that while the virus spread widely, vaccines effectively decoupled infection volume from mortality.

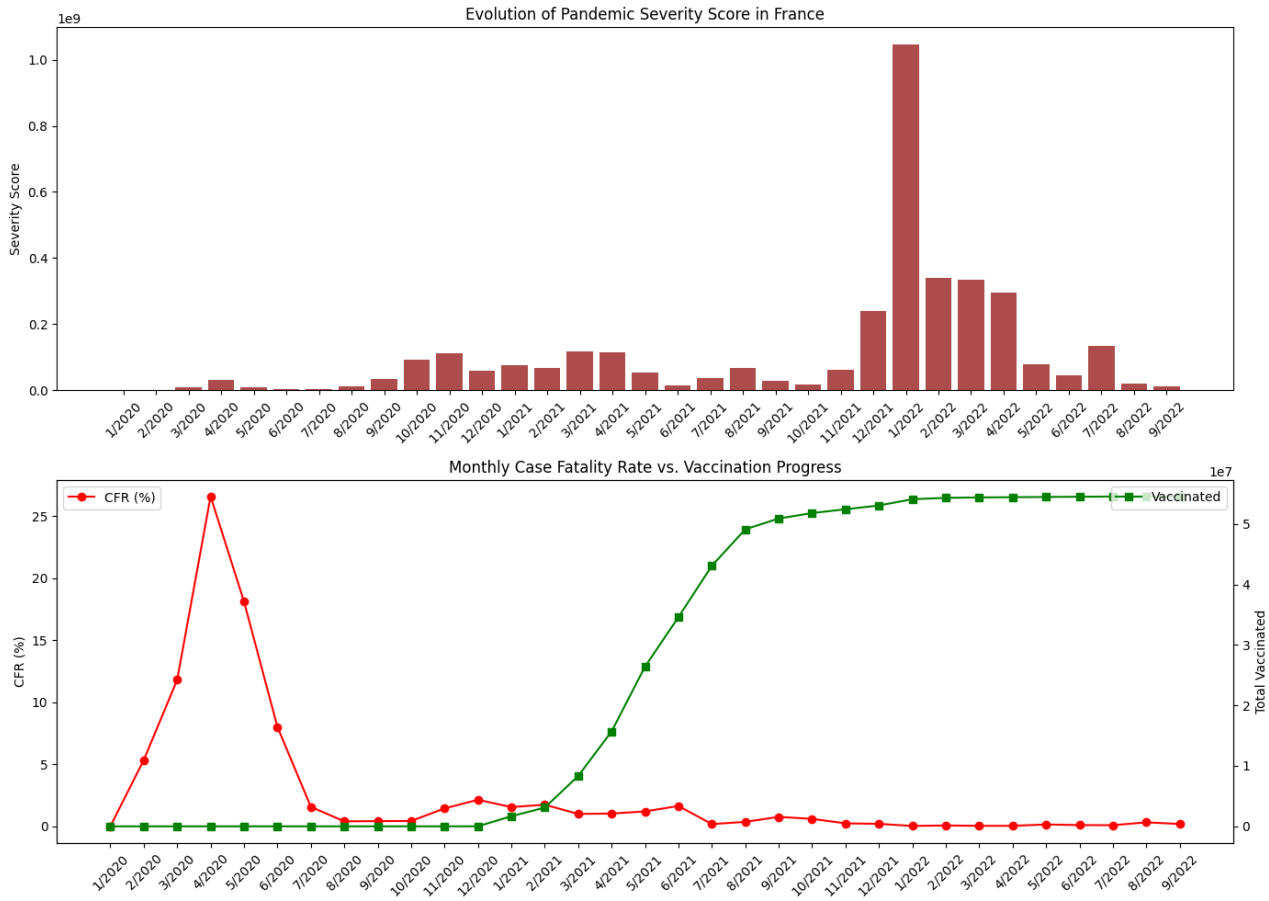

Figure 8: Top: Monthly Pandemic Severity Score in France (composite index). Bottom: Monthly CFR (%) versus cumulative vaccination progress, January 2020 – September 2022.

3 Conclusion

This project demonstrates how a well-designed relational database can be used to explore and understand complex real-world phenomena such as the COVID-19 pandemic. By structuring the data into coherent dimension and fact tables, we ensured both data integrity and analytical efficiency.

Through the different queries, we were able to investigate multiple hypotheses related to the pandemic, including the role of wealth, seasonality, vaccination, demographic structure, and healthcare capacity. The results consistently highlight that no single factor alone explains the observed outcomes. Instead, the pandemic dynamics emerge from the interaction of multiple variables, including policy responses, data quality, and structural country differences.

One key insight is the importance of proper data normalization and indicator construction. For example, using relative measures such as cases per 100,000 inhabitants or vaccination rates proved essential to avoid misleading interpretations driven by population size. Similarly, the evolution of the Case Fatality Rate over time illustrates how improvements in testing and vaccination can significantly alter the apparent severity of the disease.

Finally, this project also illustrates the importance of database systems in data science workflows. From data importation and cleaning to advanced querying and interpretation, the database serves as the backbone of the entire analytical process. As required by the project objectives, we successfully combined database design, data integration, and analytical querying to extract meaningful insights from large-scale datasets.

Future work could extend this analysis by incorporating additional variables such as mobility data, government policy indices, or hospital capacity indicators, allowing for even deeper understanding of pandemic dynamics.